Compiled Problems – Intermediate Division

1. **00-01 C3 Boolean Algebra**

   Simplify the following Boolean expression: $A(A+BC)+B(B+AC)$

2. **01-02 C2 Boolean Algebra**

   Simplify completely: $(\overline{X}+Y)(\overline{X}+\overline{Y})$

3. **01-02 C2 Boolean Algebra**

   List all ordered triples (A,B,C) that make the following expression FALSE:

   $$\overline{AB}+A(\overline{B+C})$$

4. **01-02 C3 Boolean Algebra**

   Simplify so that only AND and NOT symbols are used:

   $$\overline{A}(B+\overline{C})+\overline{B}(A+\overline{C})+\overline{C}(A+\overline{B})$$

5. **02-03 C2 Boolean Algebra**

   Simplify: (NOT A OR A AND NOT B) AND (NOT A AND B)

6. **02-03 C2 Boolean Algebra**

   Find all ordered triples that make the following expression TRUE:

   NOT A AND C OR B AND (NOT (A OR C))

7. **02-03 C3 Boolean Algebra**

   Simplify the following expression: $\overline{\overline{(A+B)}(C\overline{A})}+\overline{\overline{B}A(\overline{B}+C)}$

8. **03-04 C2 Boolean Algebra**

   Find all ordered pairs that make the following expression TRUE.

   $$\overline{AB+\overline{A}\overline{B}}$$

9. 03-04 C2 Boolean Algebra

Simplify the following expression: $B(\overline{A}+B)+A(\overline{B}+C)+\overline{C}(A+B)$

10. 03-04 C3 Boolean Algebra

Simplify the following Boolean expression: $(A+\overline{B})(\overline{AB})\ (\overline{\overline{A}+B})$

11. 04-05 C2 Boolean Algebra

Simplify: $\overrightarrow{(A+B)}\ (A\overline{B}+B\overline{C}\ )$

12. 04-05 C2 Boolean Algebra

How many ordered triples make the following expression FALSE?

$$\overline{A}(B+\overline{C})+A\overline{B}C$$

13. 04-05 C3 Boolean Algebra

Simplify the following Boolean expression so that the answer uses NOT operator and only one other type of operator:

$$\overline{A+\overline{B}}\quad\overline{C\overline{A}(B+C)}$$

14. 05-06 C2 Boolean Algebra

Simplify: $(\overline{A+\overline{B}})(A\overline{B}+C)(B+\overline{C})$

15. 05-06 C2 Boolean Algebra

How many ordered pairs make the following expression TRUE?

$$(\overline{A}+\overline{B})\overline{C}+\overline{A}(B\overline{C}+A)+(\overline{C+A})(\overline{A}B)$$

16. 05-06 C3 Boolean Algebra

Simplify the expression until only AND and NOT gates are used.

$$\overline{\overline{\overline{A}+AB}(B+C)C}$$

17. 06-07 C2 Boolean Algebra

Simplify completely: $\overline{A}(B+\overline{A})+\overline{A}(\overline{B}+A)$

18. 06-07 C2 Boolean Algebra

List all the ordered triples that make the following expression TRUE:

$$A(\overline{B}+C)+\overline{A}\,\overline{B}(A+\overline{B})C$$

19. 06-07 C3 Boolean Algebra

How many ordered triples make the following expression TRUE?

$$AB\overline{C}+A\overline{B}\,\overline{C}+\overline{A}BC+\overline{A}\,\overline{B}C$$

20. 07-08 C2 Boolean Algebra

Simplify completely: $\overline{A}(AB+C)+B(\overline{A}C+\overline{B})$

21. 07-08 C2 Boolean Algebra

What ordered triples make the following expression TRUE?

$$\overline{A}(A+B\overline{C})+B\overline{C}(A+\overline{B})$$

22. 07-08 C3 Boolean Algebra

How many ordered triples make the following expression TRUE?

$$(A\oplus B)(A+BC)$$

23. 08-09 C2 Boolean Algebra

Simplify completely: $AB\overline{\overline{(\overline{B}+C)}}+\overline{C}\,\overline{\overline{(AC+\overline{B})}}$

24. 08-09 C2 Boolean Algebra

List all ordered triples that make the following expression TRUE?

$$A\overline{(\overline{A}\overline{B}C+\overline{A}B\overline{C})}\,\overline{(\overline{A(\overline{B}C+A\overline{C})})}$$

25. 08-09 C3 Boolean Algebra

Simplify the following expression: $(A + BC)(\overline{\overline{A}B + \overline{C}})$

26. 09-10 C2 Boolean Algebra

Simplify the following Boolean expression: $(A\overline{B} + B\overline{C})(\overline{A}C + B)$

27. 09-10 C2 Boolean Algebra

Which ordered triple(s) make the following TRUE? $\overline{\overline{\overline{A(\overline{B} + C)}} + \overline{A}B + C}$

28. 09-10 C3 Boolean Algebra

Simplify the following expression: $\overline{(AB + \overline{C}) + (A(\overline{B} + C))}$

29. 10-11 C2 Boolean Algebra

List the ordered triples that make the following Boolean expression FALSE.

$$A\overline{B}(B + C) + \overline{A(\overline{\overline{B + C}})(B + \overline{C})}$$

30. 10-11 C2 Boolean Algebra

Simplify the following Boolean expression: $A(\overline{A\overline{B} + BC})$

31. 10-11 C3 Boolean Algebra

How many ordered triples make the following Boolean expression FALSE?

$$A(\overline{A}B + C) + \overline{\overline{B}(A + BC)} + C(\overline{A}B + \overline{B}C)$$

32. 11-12 C3 Boolean Algebra

Simplify the following Boolean expression: $\overline{AB} + B(\overline{A + C})$

33. 11-12 C3 Boolean Algebra

How many ordered triples make the following Boolean expression TRUE?

$$\overline{A}(B+C)+(\overline{A+C})(\overline{BC})$$

34. 12-13 C3 Boolean Algebra

Simplify the following Boolean expression: $(\overline{A}B+\overline{B})(A+B)(A\overline{B}+\overline{A})$

35. 12-13 C3 Boolean Algebra

How many ordered triples make this Boolean expression TRUE?

$$(A+\overline{B}C)(BC+\overline{A})+A(B+\overline{C})(\overline{A}+\overline{B}\overline{C})$$

36. 13-14 C3 Boolean Algebra

Which ordered triple(s) make the following Boolean expression TRUE?

$$(A\overline{B}+B\overline{C})(\overline{A}B\overline{C}+A\overline{B}C)$$

37. 13-14 C3 Boolean Algebra

How many ordered triples make this Boolean expression TRUE?

$$A\overline{B}+B(C+\overline{A})+B\overline{C}+AB\overline{C}$$

38. 14-15 C3 Boolean Algebra

Simplify the following Boolean expression:

$$(A\overline{B}+AC+B\overline{C})(B\overline{C}+\overline{A}B+\overline{A}C)$$

39. 14-15 C3 Boolean Algebra

How many ordered triples make this Boolean expression FALSE?

$$\overline{AB}(\overline{\overline{A}+B+C})(\overline{A+B\overline{C}})$$

40. 15-16 C3 Boolean Algebra

Which ordered triple(s) make the following Boolean expression TRUE?

$$\overline{A}(B + A\overline{C}) + B(C + \overline{A}B)$$

41. 15-16 C3 Boolean Algebra

How many ordered triples make the following Boolean expression TRUE?

$$\overline{A}(\overline{B + A\overline{C}}) + \overline{BC}(\overline{\overline{A} + C})$$